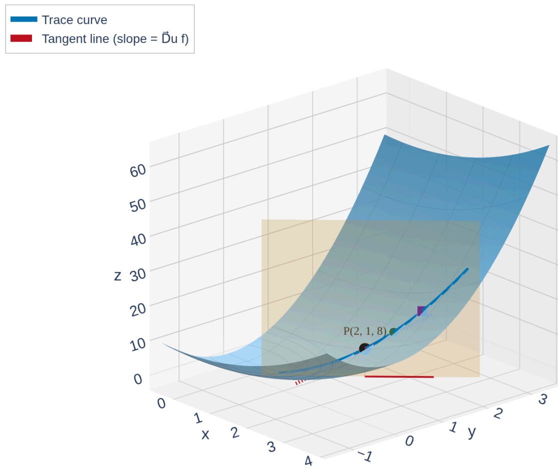
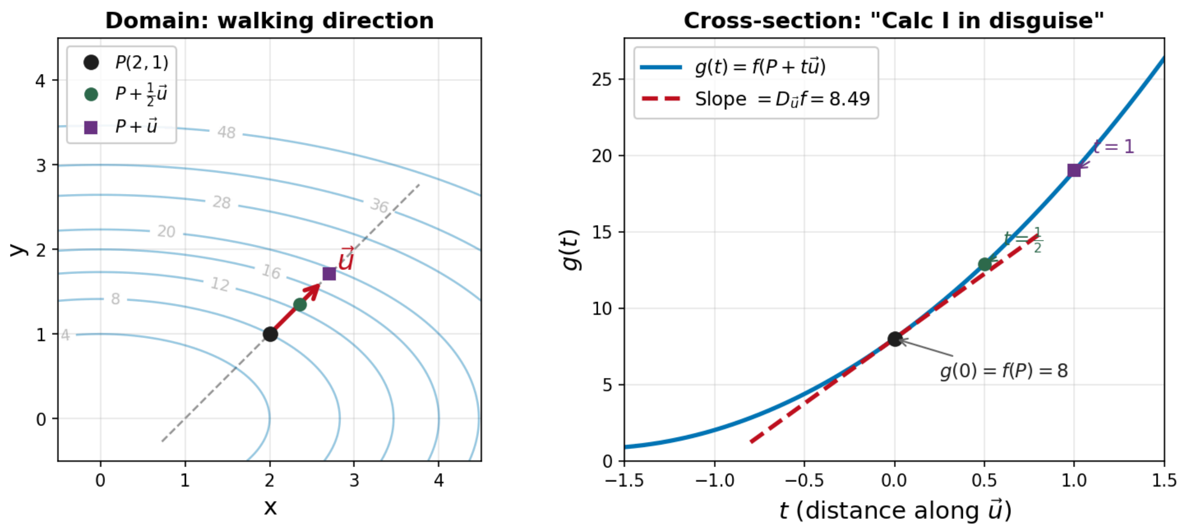
Section 14.6: Directional Derivatives and the Gradient Vector

Partial Derivatives

Directional Derivatives — Motivation



Interactive version



The directional derivative is just a **Calc I derivative in disguise**: pick a direction, slice the surface, and find the slope.

Definition of the Directional Derivative

Directional Derivative

Let $\vec{u} = \langle a, b \rangle$ be a **unit vector** ($|\vec{u}| = 1$). The **directional derivative** of f at (x, y) in the direction of $\vec{u}$ is:

$$D_{\vec{u}} f(x, y) = \lim_{t \to 0} \frac{f(x + ta,\ y + tb) - f(x, y)}{t}$$

provided this limit exists.

This is the same **rise-over-run** idea from Calc I, but now we walk in the direction $\vec{u}$ instead of along the x-axis.

Example 1: $D_{\vec{u}} f$ from the Definition

Example 1

Let $f(x, y) = x^2 + 4y^2$ and $\vec{u} = \left\langle \frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2} \right\rangle$.

Find $D_{\vec{u}} f(2, 1)$ using the **limit definition**.

The Gradient Vector

Gradient Vector

If f has partial derivatives f_x and f_y, the **gradient** of f is the vector:

$$\nabla f(x, y) = \langle f_x(x, y),\ f_y(x, y) \rangle = f_x \vec{i} + f_y \vec{j}$$

The symbol ∇ is called **nabla** or **del**.

Example 2

Find ∇f for $f(x, y) = x^2 + 4y^2$. Then evaluate $\nabla f(2, 1)$.

The Big Theorem: $D_{\vec{u}} f = \nabla f \cdot \vec{u}$

Theorem: Directional Derivative via the Gradient

If f is a differentiable function of x and y, then:

$$D_{\vec{u}} f(x, y) = \nabla f(x, y) \cdot \vec{u}$$

for any unit vector $\vec{u}$.

This is the **central formula** of Section 14.6. It connects direction, rate of change, and gradient in one equation.

The dot product appears **because of the chain rule**. Each partial derivative gets multiplied by the corresponding component of $\vec{u}$, which is exactly what a dot product does.

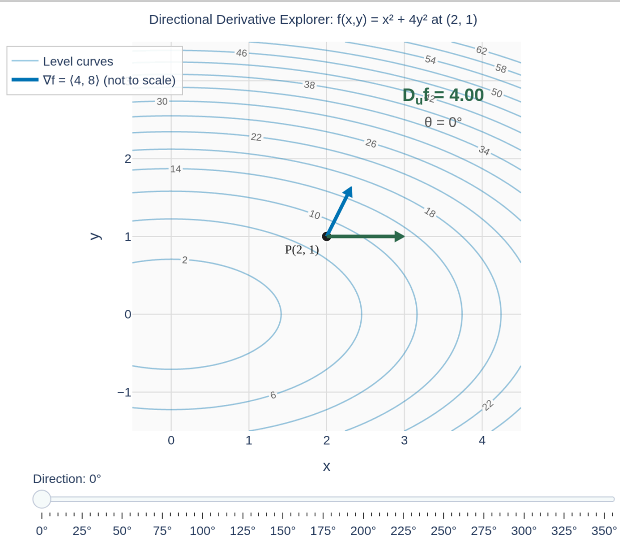
Example 3: $D_{\vec{u}}f$ via the Gradient

Example 3

Same problem as Example 1: $f(x, y) = x^2 + 4y^2$, $\vec{u} = \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$.

Find $D_{\vec{u}}f(2, 1)$ using the gradient.

Maximizing the Directional Derivative



Interactive version

Notice: $D_{\vec{u}}f$ is **largest** when $\vec{u}$ aligns with ∇f, **most negative** when opposing, and **zero** when perpendicular. Why?

The gradient tells you everything:

- Its **direction** = steepest ascent
- Its **magnitude** $|\nabla f|$ = the maximum rate of change
- **Perpendicular** to it = no change (along the level curve)

Example 4: Fastest Increase and Decrease

Example 4

For $f(x, y) = x^2 + 4y^2$ at the point $(2, 1)$:

(a) What direction gives the **fastest increase** in f? How fast?

(b) What direction gives the **fastest decrease**?

(c) In what directions is $D_{\vec{u}} f = 0$?

Example 5: Which Way Should I Walk?

Example 5

A hiker is at position $(2, 1)$ on a mountain whose elevation is modeled by $f(x, y) = x^2 + 4y^2$ (hundreds of feet).

(a) She wants to **descend as steeply as possible**. What direction should she walk?

(b) She wants to **stay at the same elevation** (walk along a contour line). What direction(s) should she walk?

Your Turn: Directional Derivative Practice

Example 6

Let $f(x, y) = xe^{-y}$ and let $\vec{u}$ be the unit vector in the direction of $\langle 2, -1 \rangle$.

(a) Find $\nabla f(2, 0)$.

(b) Find $D_{\vec{u}} f(2, 0)$.

(c) What is the maximum rate of change of f at $(2, 0)$?

Gradient Is Perpendicular to Level Curves

Theorem: Gradient Perpendicular to Level Curves

The gradient vector $\nabla f(a, b)$ is **perpendicular** to the level curve $f(x, y) = k$ at the point (a, b).

The gradient always points **perpendicular to level curves**, aiming toward higher values of f. This is why contour maps and gradients are deeply connected.

Example 7: Tangent Line to a Level Curve

Example 7

Consider $f(x, y) = x^2 + 4y^2$ with level curves that are ellipses.

At the point $(2, 1)$, find the equation of the **tangent line** to the level curve $f(x, y) = 8$.

Extending to Three Variables

Gradient in Three Variables

$$\nabla f(x, y, z) = \langle f_x,\ f_y,\ f_z \rangle$$

Since ∇F is perpendicular to the level surface, it serves as the **normal vector** for the tangent plane!

Tangent plane to $F(x, y, z) = k$ at (a, b, c):

$$F_x(a, b, c)(x - a) + F_y(a, b, c)(y - b) + F_z(a, b, c)(z - c) = 0$$

Example 8: Tangent Plane to a Level Surface

Example 8

Let $F(x, y, z) = zx^2 - yz^2$.

(a) Compute $\nabla F(1, 2, 3)$.

(b) Find the equation of the tangent plane to the level surface $F(x, y, z) = -15$ at the point $(1, 2, 3)$.

Homework

Section 14.6: 1, 5, 9, 11, 13, 19, 21, 29, 31, 35, 39, 44, 49, 51, 55, 61, 65, 73